

Introduction to Database Management Systems

CS4604

ENTITY-RELATIONSHIP
DIAGRAM REPORT

AI-Powered Landscaping Cost Estimation and Contractor Platform

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2. Problem Definition

The proposed system is an AI-powered landscaping platform designed to help customers understand the likely cost and scope of yard projects before hiring a contractor. Customers provide property information, submit a landscaping request, and upload yard photos. The system uses this information to produce an AI-generated preliminary estimate, including an estimated project size and price. Customers can then receive quotes from landscaping contractors, schedule appointments, track the resulting job, receive invoices, and record payments. The database must preserve the information needed to support this workflow while clearly connecting customers, properties, requests, AI estimates, contractors, quotes, appointments, jobs, invoices, payments, and subscription plans.

Database design objective. The ER model organizes the major real-world objects of the platform and defines how they relate so the later relational tables can be created with clear keys, constraints, and data ownership.

3. System Users and Privileges

System User	Privileges
Customer	Create and manage an account; maintain contact information; add property information; submit landscaping requests; upload yard photos; view AI-generated estimates; review contractor quotes; schedule appointments; track jobs; view invoices; make payments; and select a subscription plan.
Contractor / Landscaper	Maintain contractor and business information; review relevant project requests; submit and update quotes; participate in scheduled appointments; perform accepted jobs; and update job progress, cost, and completion information.
System Administrator	Manage system users and platform data; maintain landscaping service definitions and subscription plans; review project, quote, job, invoice, and payment records when needed for system administration; and maintain appropriate access to system data.

4. Entities and Brief Descriptions

The entity descriptions below explain what each entity represents in the database. Attributes are intentionally omitted from this section because they are shown in the ER diagram.

Entity	Description
System Admin	Represents an administrator responsible for managing the platform and maintaining system-level data.
Customer	Represents a homeowner or other customer who uses the platform to request landscaping work.
Property	Represents a physical property or yard owned or managed by a customer where landscaping work may be requested.
Landscaping Service	Represents a type of landscaping work offered through the platform, such as lawn care, planting, cleanup, or related services.

Entity	Description
Project Request	Represents a customer request describing landscaping work needed for a specific property and service.
Photo	Represents an image uploaded for a project request to provide visual information about the yard or work area.
AI Estimate	Represents the preliminary project estimate produced by the AI system from the information associated with a project request.
Subscription Plan	Represents a platform membership or service tier available to customers.
Contractor	Represents a landscaper or landscaping business that can respond to customer project requests.
Contractor Quote	Represents a contractor's proposed scope and final quoted price for a specific customer project request.
Appointment	Represents a scheduled meeting, site visit, or service appointment involving a customer, contractor, and project request.
Job	Represents the landscaping work created after a contractor quote is accepted and work proceeds.
Invoice	Represents the formal amount billed for a completed or billable landscaping job.
Payment	Represents money received against an invoice, including the amount, method, date, and payment status.

5. Relationships and Brief Descriptions

Relationship names are written as actions so that each connection can be read as a simple statement. Cardinality indicates how many records of one entity can be associated with records of the other entity.

Relationship	Description	Card.
Customer owns Property	A customer can own or manage multiple properties, while each property belongs to one customer.	1:M
Customer submits Project Request	A customer can submit multiple project requests, while each request is submitted by one customer.	1:M
Property has Project Request	A property can have multiple landscaping requests over time, while each request is associated with one property.	1:M
Landscaping Service is requested by Project Request	One landscaping service can be selected by many project requests, while each project request identifies one primary service.	1:M

Relationship	Description	Card.
Project Request includes Photo	A project request can contain multiple yard photos, while each photo belongs to one request.	1:M
Subscription Plan is selected by Customer	A subscription plan can be selected by many customers, while a customer has at most one active plan in this model.	1:M
Project Request generates AI Estimate	Each project request produces one preliminary AI estimate for the project.	1:1
Project Request receives Contractor Quote	A project request may receive quotes from multiple contractors, while each quote applies to one request.	1:M
Contractor submits Contractor Quote	A contractor can submit multiple quotes, while each quote is submitted by one contractor.	1:M
AI Estimate is referenced by Contractor Quote	The AI estimate may be used as a reference when one or more contractor quotes are prepared.	1:M
Customer schedules Appointment	A customer may have multiple appointments, while each appointment belongs to one customer.	1:M
Contractor attends Appointment	A contractor may have multiple appointments, while each appointment is assigned to one contractor.	1:M
Project Request has Appointment	A project request may have one or more appointments such as site visits or scheduled work.	1:M
Contractor Quote is accepted as Job	An accepted quote can create one job; a quote that is not accepted creates no job.	1:0..1
Contractor performs Job	A contractor can perform multiple jobs, while each job is assigned to one contractor.	1:M
Job generates Invoice	A job generates one invoice in this model, and each invoice belongs to one job.	1:1
Invoice receives Payment	An invoice can receive one or more payments, allowing the model to support full or partial payment.	1:M
System Admin manages Landscaping Service	The system administrator maintains the service types available on the platform.	Admin
System Admin manages Subscription Plan	The system administrator maintains the available subscription plans and tiers.	Admin

Design Reference

This report follows standard ER modeling conventions in which an ERD shows entities, attributes, relationships, and relationship cardinality. Primary keys uniquely identify entity instances, while foreign keys express connections between related records. Source: [IBM - Entity Relationship Diagram](https://www.ibm.com/think/topics/entity-relationship-diagram) (https://www.ibm.com/think/topics/entity-relationship-diagram).

6. ER Diagram

The diagram below preserves the original system entities while presenting the attributes, primary/foreign keys, relationship names, and cardinalities in a cleaner format.

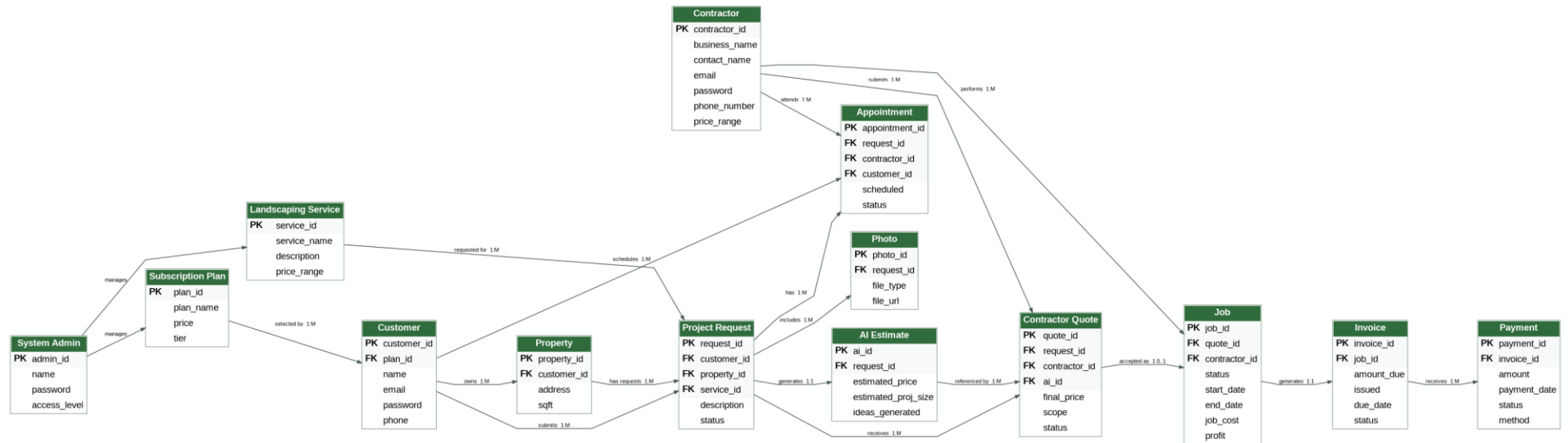


Figure 1. AI-Powered Landscaping Platform - Entity-Relationship Diagram